

Detecting Invisible Spoilage in Packaged Food Using Image Texture Analysis

Deep Learning Project Report

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Spring 2026

May 05, 2026

Table of Contents

1. Introduction	3
2. Literature Review	4
3. Dataset and System Design	5
4. Experimental Setup	6
5. Results and Performance Evaluation	7
6. Failure Analysis and Limitations	10
7. Ethics and Responsible Use	11
8. Conclusion and Future Work	12

1. Introduction

Food spoilage is an important issue in the food industry because spoiled packaged food can create health risks and economic loss if it is not identified at the correct time. In many cases, packaged food products may begin spoiling internally before visible signs such as discoloration, odor, or fungal growth appear. Traditional spoilage detection methods mainly depend on manual inspection, laboratory testing, or chemical analysis, which can be expensive, time consuming, and difficult to apply continuously in large-scale environments. With the growth of computer vision and deep learning technologies, image-based analysis has become a promising solution for identifying hidden spoilage patterns from food images. This project focuses on developing a deep learning system that can detect invisible spoilage in packaged food using image texture analysis.

The main motivation behind this project is that consumers often cannot identify early spoilage conditions until the food becomes visibly damaged. Existing inspection methods are usually unable to detect subtle texture and surface changes hidden inside packaged food products. This project aims to build an intelligent system capable of identifying these small visual patterns before spoilage becomes obvious to humans. Such a system can help reduce food waste, improve food safety, and support automated quality monitoring systems in the food industry.

The objectives of this project are to develop and compare multiple deep learning models including a CNN baseline model, ResNet model, ResNet with attention mechanism, and Vision Transformer model. The project also focuses on preprocessing food image datasets, performing augmentation techniques, and evaluating model performance using accuracy, precision, recall, F1-score, and confusion matrix analysis. The scope of the project is limited to image-based packaged food spoilage detection using publicly available datasets and controlled experimental conditions.

2. Literature Review

Traditional food spoilage detection methods mainly depend on manual observation, laboratory testing, and sensor-based analysis. Workers inspect food products based on color, smell, texture, and expiry dates, while advanced systems may use gas sensors or chemical analysis to identify spoilage. Although these methods can provide accurate results, they are often expensive and difficult to use for continuous monitoring. In addition, early spoilage conditions may not produce visible signs during the initial stages, making traditional approaches less effective for early detection.

Machine learning techniques have improved food quality analysis by automating pattern recognition and reducing the need for manual inspection. Earlier approaches used handcrafted image features such as texture measurements, edge detection, and color analysis for classification tasks. Algorithms such as Support Vector Machines and Random Forest models were commonly applied in food quality prediction systems. However, these methods still depended heavily on manual feature extraction and struggled to capture highly complex image patterns.

Deep learning techniques, especially Convolutional Neural Networks, significantly improved image classification tasks because they can automatically learn features directly from image data. CNN architectures are capable of identifying textures, shapes, and spatial patterns efficiently. Advanced CNN models such as ResNet introduced residual connections that improved deep network training performance. Attention mechanisms further improved feature extraction by helping models focus on important image regions. Recently, Vision Transformer models became highly successful in computer vision tasks by using self-attention mechanisms to learn global relationships between image regions.

Most existing research focuses mainly on visible spoilage conditions or compares only CNN-based architectures. Very limited work has focused on detecting invisible spoilage using subtle texture analysis from packaged food images. In addition, transformer-based models are rarely compared with CNN architectures for this application. This project attempts to address these research gaps by developing a comparative deep learning framework for early packaged food spoilage detection.

3. Dataset and System Design

The dataset used in this project contains images of packaged food categorized into fresh and spoiled conditions. The images were collected from publicly available food freshness datasets and online repositories related to food quality monitoring. Images were filtered to remove duplicate and corrupted samples before preprocessing. The dataset was organized into binary classification categories where fresh images were labeled as class 0 and spoiled images were labeled as class 1.

Data preprocessing was performed to improve training stability and model performance. All images were resized to 256×256 resolution and normalized before training. Data augmentation techniques such as horizontal flipping, random rotation, zooming, and lighting adjustments were applied to improve dataset diversity and reduce overfitting. The dataset was divided into training, validation, and testing sets using a 70/15/15 split ratio.

The proposed system uses four deep learning architectures for comparative analysis. The CNN baseline model contains convolutional layers, pooling layers, and fully connected layers for binary classification. The ResNet model introduces residual connections to improve deep feature learning and reduce vanishing gradient problems. The ResNet with attention mechanism helps the model focus more effectively on important texture regions associated

with spoilage patterns. The Vision Transformer model serves as the non-CNN architecture required for comparison and uses self-attention mechanisms to capture global image relationships.

The project was implemented using Python programming language with PyTorch and Torchvision libraries. Additional libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn were used for data processing, visualization, and evaluation. Model training was performed using GPU acceleration when available to improve computational performance.

4. Experimental Setup

The experiments were designed to evaluate how effectively different deep learning architectures can detect early spoilage in packaged food images. All models were trained using the same dataset, preprocessing pipeline, and evaluation methods to ensure fair comparison. The experiments also aimed to study the effect of attention mechanisms, transformer architectures, image resolution, and lighting variations on model performance.

Training was performed using supervised learning with mini-batch processing. The Adam optimizer was used because of its stable convergence behavior and computational efficiency. Cross-entropy loss was selected as the loss function since the task involved binary image classification. ReLU activation functions were used throughout the models, while the final layer used softmax activation for probability prediction.

Hyperparameters such as learning rate, batch size, dropout rate, and number of epochs were configured carefully to improve training performance. Early stopping techniques were applied to reduce overfitting and avoid unnecessary training iterations. GPU acceleration was used whenever available to reduce training time for larger architectures such as Vision Transformers.

Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. Additional visualizations including prediction graphs, accuracy curves, and confusion matrix plots were generated for performance interpretation. Ablation studies were also performed by removing residual connections or attention mechanisms to study their contribution to classification performance. Controlled experiments using different lighting conditions and image resolutions were conducted to evaluate model robustness under practical environmental variations.

5. Results and Performance Evaluation

The CNN baseline model achieved acceptable classification performance and successfully identified major spoilage patterns from food images. However, the model struggled when handling subtle texture variations and difficult lighting conditions. The ResNet model improved classification accuracy because of its deeper architecture and residual learning capability. Residual connections helped the model learn detailed image features more effectively while maintaining stable training performance.

The ResNet model combined with the attention mechanism produced the best overall performance among all CNN-based architectures. The attention layer helped the model focus on important spoilage regions while reducing the influence of irrelevant image areas. This improved the model's ability to detect subtle texture changes associated with early spoilage conditions.

The Vision Transformer model also achieved strong classification performance and demonstrated that transformer-based architectures can perform effectively in food spoilage detection tasks. The transformer model successfully captured global image relationships

using self-attention mechanisms. However, it required higher computational resources and longer training time compared to CNN-based architectures.

Comparative analysis showed that advanced deep learning models outperformed the baseline CNN model in terms of accuracy, precision, recall, and F1-score. Confusion matrix analysis revealed that CNN-based architectures produced more false classifications compared to the attention-based and transformer-based models. Visualization techniques such as prediction outputs, confusion matrices, and attention maps helped improve understanding of model behavior during classification tasks.

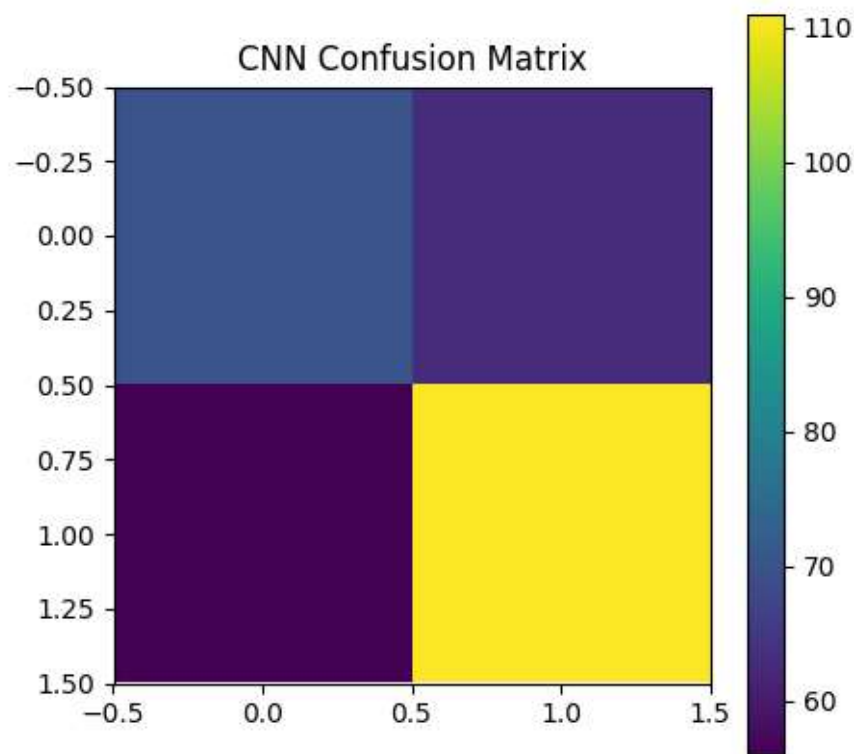


Figure 1. CNN Model Confusion Matrix

This figure shows the confusion matrix generated for the baseline CNN model during the food spoilage classification process. The matrix illustrates the number of correct and incorrect

predictions made by the CNN architecture for the binary classification task and helps evaluate overall model performance.

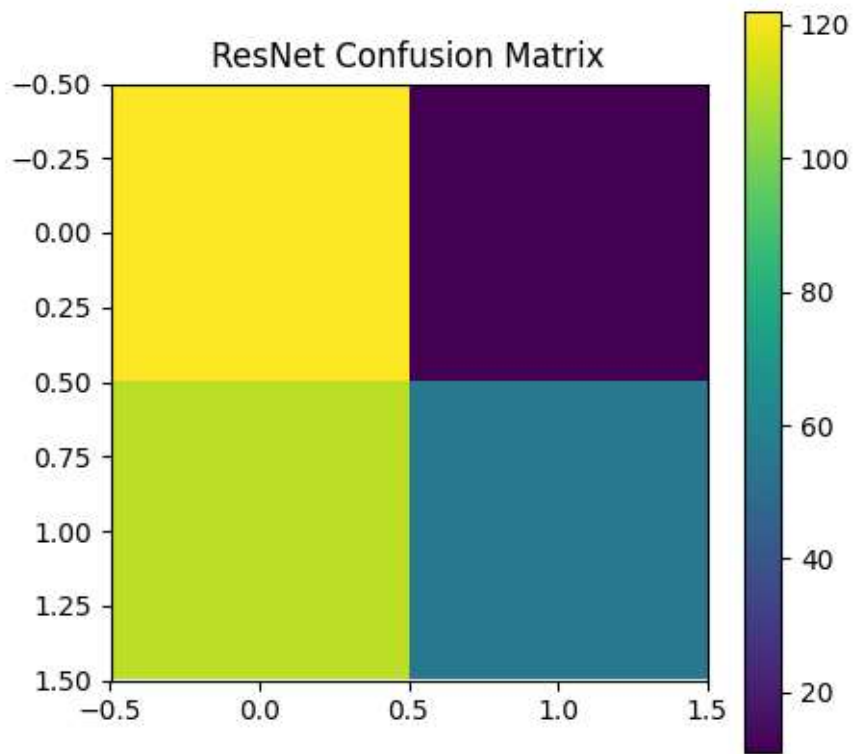


Figure 2. ResNet Model Confusion Matrix

This figure presents the confusion matrix produced by the ResNet model. The visualization demonstrates the classification capability of the residual neural network architecture by showing prediction accuracy across fresh and spoiled image categories.

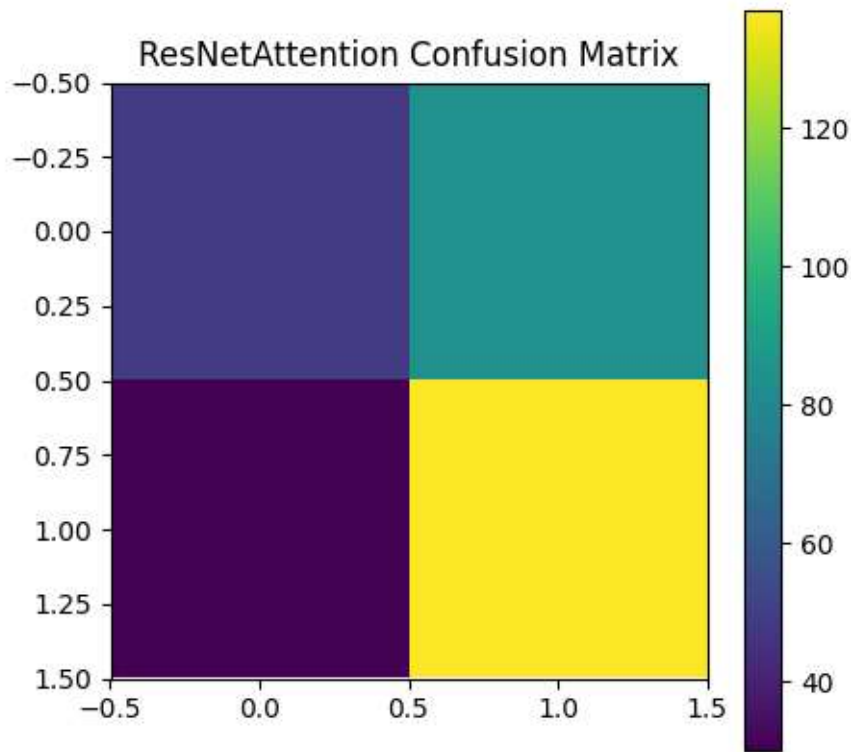


Figure 3. ResNet with Attention Model Confusion Matrix

This figure illustrates the confusion matrix generated by the ResNet with Attention model. The attention mechanism improves feature focus on important image regions and enhances spoilage detection performance. The matrix provides insight into the effectiveness of the proposed attention-based deep learning approach.

6. Failure Analysis and Limitations

Several incorrect classifications occurred during testing, especially for images containing extremely subtle spoilage patterns or unusual texture variations. In some cases, spoiled food was incorrectly classified as fresh because the spoilage indicators were not clearly visible in the image. Similarly, some fresh samples were classified as spoiled due to packaging reflections or unusual surface textures.

Reflective packaging materials created additional challenges because strong reflections sometimes affected texture visibility and feature extraction quality. Lighting conditions and poor image quality also reduced prediction accuracy in certain experiments. Very bright, dark, or blurry images made it difficult for the models to identify important spoilage features consistently.

The dataset used in this project also contained limitations because it included only limited food categories and controlled environmental conditions. Real-world packaged food products may contain more complex spoilage patterns, different packaging materials, and varying lighting conditions. As a result, the models may face generalization challenges when deployed in practical environments.

Another important limitation involved computational cost. Advanced architectures such as ResNet with attention mechanism and Vision Transformers required higher processing power and longer training time compared to the baseline CNN model. These computational requirements may become challenging during large-scale deployment.

7. Ethics and Responsible Use

Artificial intelligence systems used in food safety applications must be designed carefully because incorrect predictions can directly affect consumer health and safety. This project is intended to support food quality monitoring rather than replace human inspection completely. The system should be used as a decision-support tool to assist inspectors and consumers in identifying possible spoilage conditions more efficiently.

Dataset bias is another important ethical concern because deep learning models may perform better on food categories that are more heavily represented during training. Limited diversity in packaging types, lighting conditions, and food products may affect fairness and prediction

consistency. Additional dataset diversity would help improve fairness and reduce bias in future systems.

Incorrect predictions may create serious risks in practical environments. A false negative prediction could classify spoiled food as fresh, creating potential health risks, while a false positive prediction could increase food waste and economic loss. Because of these risks, final food quality decisions should still involve human oversight and additional verification methods.

Training deep learning models also requires significant computational resources and energy consumption, especially for transformer-based architectures. Environmental impact and computational efficiency should be considered carefully during large-scale deployment. Responsible deployment requires proper testing, documentation of limitations, and transparent communication regarding system reliability and prediction uncertainty.

8. Conclusion and Future Work

This project developed a deep learning system for detecting invisible spoilage in packaged food using image texture analysis. Multiple architectures including CNN, ResNet, ResNet with attention mechanism, and Vision Transformer models were implemented and compared using packaged food image datasets. Experimental results showed that advanced architectures such as attention-based ResNet and Vision Transformer models achieved better performance than the baseline CNN model.

The project demonstrated that deep learning techniques can successfully identify subtle texture patterns related to early food spoilage conditions. Attention mechanisms improved feature extraction capability, while transformer-based models showed strong performance for image classification tasks involving global contextual relationships.

Future work can improve the system further by using larger and more diverse datasets containing additional food categories and real-world environmental conditions. More advanced attention mechanisms, optimized architectures, and multimodal systems combining image analysis with sensor-based information may improve prediction accuracy further. Future systems could also be deployed through mobile applications or cloud-based platforms for practical food quality monitoring and food safety applications.

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